Linear Algebra and Analytic Geometry

Unit 1: Matrices and Systems

Worked Solutions to Exercises 1–34

Alex Vergara

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1. Consider the matrix

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}.$$

Verify that there exists a power of A which is the zero matrix.

Solution:

Given, $A \in \mathcal{M}_3(\mathbb{R})$ they are asking for:

$$\exists k \in \mathbb{N}+ : A^k = 0_3.$$

Computing A^2:

$$\begin{aligned} A^2 &= AA \\ &= \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix} \neq 0_3. \end{aligned}$$

Computing A^3:

$$\begin{aligned} A^3 &= A^2 A \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} \\ &= \boxed{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}. \end{aligned}$$

Therefore,

$$\boxed{\exists k = 3 \in \mathbb{N}^+ : A^k = 0_3}.$$

2. Consider the matrices

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

Compute $(AB)^2$, A^2B^2, and B^2A^2.

Solution:

Given $A, B \in \mathcal{M}_{2\times 2}(\mathbb{R})$, all three products are defined.

Product $(AB)^2$:

First, compute AB:

$$\begin{aligned} AB &= \begin{bmatrix} 1(1)+1(-1) & 1(1)+1(1) \\ 1(1)+0(-1) & 1(1)+0(1) \end{bmatrix} \\ &= \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix}. \end{aligned}$$

Therefore,

$$\begin{aligned} (AB)^2 &= \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix} \\ &= \boxed{\begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}}. \end{aligned}$$

Product A^2B^2:

Computing both squares:

$$A^2 = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix},$$

$$B^2 = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}.$$

Then,

$$\begin{aligned} A^2B^2 &= \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} \\ &= \boxed{\begin{bmatrix} -2 & 4 \\ -2 & 2 \end{bmatrix}}. \end{aligned}$$

Product B^2A^2:

$$B^2A^2 = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

$$= \boxed{\begin{bmatrix} 2 & 2 \\ -4 & -2 \end{bmatrix}}.$$

Hence,

$$\boxed{(AB)^2 \neq A^2B^2 \neq B^2A^2}.$$

In general, $(AB)^2 = ABAB$, which cannot be rearranged as A^2B^2 without assuming $AB = BA$.

3. State, with justification, whether the following statements are true or false.

   a) For any square matrices A and B of the same order, $AB = BA$.

   b) For any square matrices A and B of the same order, if $AB = 0$, then $A = 0$ or $B = 0$.

   c) For any square matrices A, B, and C of the same order, if $AB = AC$, then $B = C$.

   d) If, for a square matrix A, there exists $k \in \mathbb{N}$ such that $A^k = 0$, then $A = 0$.

   e) For any square matrices A and B of the same order, $(AB)^2 = A^2B^2$.

4. Final item. Remove when done.